# Software Requirements Specification

## Procurement Process Digitalization — MVP

| Field | Details |
|---|---|
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| **Prepared by** | Ghaith Hasan |
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## Revision History

| Version | Date | Author | Description |
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| 0.1 | 2026-03-08 | Ghaith Hasan | Initial draft |
| 1.0 | 2026-03-16 | Ghaith Hasan | First complete version |

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# 1. Introduction

## 1.1 Purpose

This document defines the Software Requirements Specification (SRS) for the **Procurement Process Digitalization MVP**, developed for a manufacturing factory as part of the AVERROA IT Launchpad program. It serves as the authoritative reference for design, development, testing, and stakeholder review, following the IEEE Std 830-1998 standard.

## 1.2 Scope

**System Name:** ProcureFlow MVP

The system digitalizes the first two steps of the procurement cycle: Purchase Request (PR) creation and Request for Quotation (RFQ) PDF generation.

**In Scope:**

- Digital PR creation with structured form and auto-numbering.
- Automated RFQ PDF generation based on approved PR data.
- Master data registry for products, raw materials, BOMs, and vendors.
- Role-based user access control.

**Out of Scope (Future Phases):**

- Automated email dispatch to vendors.
- Quotation collection and comparison.
- Purchase Order creation.
- ERP integration.

## 1.3 Definitions and Acronyms

| Term | Definition |
|---|---|
| PR | Purchase Request — internal request to procure raw materials |

| Term | Definition |
|---|---|
| RFQ | Request for Quotation — formal document sent to vendors requesting pricing |
| PO | Purchase Order — legally binding order issued to a selected vendor |
| BOM | Bill of Materials — list of raw materials required to produce a product |
| MVP | Minimum Viable Product |
| UoM | Unit of Measure (e.g., kg, litre, piece) |
| RBAC | Role-Based Access Control |
| ORM | Object-Relational Mapping |

## 1.4 References

| Reference | Description |
|---|---|
| IEEE Std 830-1998 | Recommended Practice for Software Requirements Specifications |
| Assignment 2 Brief | AVERROA Procurement Process Digitalization assignment |
| ISO 9001:2015 | Quality Management — Procurement control requirements |

# 2. Overall Description

## 2.1 Product Perspective

ProcureFlow MVP is a standalone web application and the first digitalization step in a broader procurement transformation. It does not replace an ERP system at this stage but is architected to integrate with one in the future through a REST API layer. The system sits between the Planning team (who initiate PRs) and the Purchasing team (who communicate with vendors).

## 2.2 Current Process and Pain Points

The factory's procurement cycle today is fully manual: planning creates paper-based Purchase Requests, purchasing manually drafts emails to three vendors, collects quotes in a spreadsheet, selects the lowest price, and types the Purchase Order. This results in the following pain points:

| # | Pain Point | Business Impact |
|---|---|---|
| 1 | No structured PR form | Missing fields, inconsistent data, errors in orders |
| 2 | No document numbering | Cannot trace or reference PRs in audits or communication |
| 3 | Manual RFQ drafting | 15–30 minutes of repetitive effort per quotation cycle |
| 4 | No central data repository | Procurement history lost; no analytics possible |
| 5 | No approval visibility | Management cannot see pending requests in real time |

## 2.3 User Classes and Characteristics

| Role | Description | Access Level |
|---|---|---|
| Planner | Creates PRs based on production schedule | Create and view own PRs |
| Purchasing Officer | Reviews PRs, generates and downloads RFQ PDFs | Full PR and RFQ management |
| Procurement Manager | Approves or rejects PRs before RFQ release | View all records, approve PRs |
| System Administrator | Manages users and master data | Full system access |

## 2.4 Operating Environment

- **Client:** Modern web browser (Chrome 110+, Firefox 110+, Edge 110+) on desktop or laptop.
- **Deployment:** On-premise local server for MVP; cloud-ready for future phases.
- **Database:** SQLite for development; PostgreSQL for production.
- **OS:** Ubuntu 20.04+ or Windows Server 2019+.

## 2.5 Constraints and Assumptions

- Technology stack must be open-source or free-tier to avoid licensing costs at MVP stage.
- The generated PDF must be professional enough to be sent directly to external vendors without modification.
- Users have basic computer literacy sufficient to use a standard web form.
- A list of current vendors, products, and raw materials will be provided for initial data seeding.

# 3. External Interface Requirements

## 3.1 User Interface

- **Dashboard:** Summary view showing PR counts by status.
- **PR Form:** Structured form with validated dropdown fields, date picker, and free-text notes.
- **PR List View:** Filterable and searchable table with status badges and action buttons.
- **RFQ Preview:** Content preview before PDF export.
- **Admin Panel:** CRUD interface for products, materials, BOM entries, and vendors.

All forms shall include client-side validation with clear error messages. Status badges shall be color-coded: Draft (grey), Pending (orange), Approved (green), Cancelled (red).

## 3.2 Software Interfaces

| Layer | Technology | Purpose |
|---|---|---|
| Web Framework | Python / Flask | Application routing and business logic |
| ORM | SQLAlchemy | Database abstraction layer |
| Frontend | Jinja2 + Bootstrap 5 | Server-side templating and responsive UI |
| PDF Engine | WeasyPrint | HTML/CSS template rendering to PDF |
| Database | SQLite (dev) / PostgreSQL (prod) | Data persistence |
| Authentication | Flask-Login + bcrypt | Session management and password security |

## 3.3 Communication Interfaces

- HTTP/HTTPS for all browser-to-server communication.
- PDF is downloaded via browser and manually attached to email by the user.
- RESTful API endpoints will be exposed in future phases for ERP integration.

# 4. Functional Requirements

## 4.1 Purchase Request Management

| ID | Requirement | Priority |
|---|---|---|
| FR-PR-001 | The system shall auto-generate a unique PR number in the format `PR-YYYY-NNNN` upon submission. | Critical |
| FR-PR-002 | The PR form shall capture: Product Name, Raw Material, Quantity, Unit of Measure, Required Date, Requester (auto from login), Department (optional), and Notes (optional). | Critical |
| FR-PR-003 | The system shall validate that Quantity is a positive number greater than zero. | Critical |
| FR-PR-004 | The system shall validate that Required Date is not in the past. | High |
| FR-PR-005 | A newly created PR shall be saved with status `Draft` by default. | Critical |
| FR-PR-006 | The Procurement Manager shall be able to transition PR status to `Pending`, `Approved`, or `Cancelled`. | High |
| FR-PR-007 | The PR list shall display: PR Number, Product, Material, Quantity, UoM, Required Date, Requester, Status, and Actions. | High |
| FR-PR-008 | The PR list shall support filtering by Status and searching by PR Number or Material Name. | High |
| FR-PR-009 | A PR in `Approved` status shall be locked from editing; only cancellation shall be permitted. | High |

| ID | Requirement | Priority |
|---|---|---|
| FR-PR-010 | The system shall record creation timestamp, last update timestamp, and status change history for every PR. | Medium |

## 4.2 RFQ Generation and PDF Export

| ID | Requirement | Priority |
|---|---|---|
| FR-RFQ-001 | The system shall allow RFQ generation only from a PR with status `Approved`. | Critical |
| FR-RFQ-002 | The system shall auto-generate a unique RFQ number in the format `RFQ-YYYY-NNNN`. | Critical |
| FR-RFQ-003 | The generated RFQ PDF shall include: Company name and logo, RFQ Number, Issue Date, PR Reference, Material Name and Specification, Quantity, UoM, Required Delivery Date, Quotation Submission Deadline, Vendor instructions, Company contact details, and a signature placeholder. | Critical |
| FR-RFQ-004 | The quotation submission deadline shall be auto-calculated as Issue Date plus a configurable number of days (default: 7). | High |
| FR-RFQ-005 | The PDF file shall be downloadable directly via the browser upon generation. | Critical |
| FR-RFQ-006 | Each PR shall produce one RFQ record. Re-generation shall increment a version counter. | High |
| FR-RFQ-007 | The PDF filename shall follow the format: `RFQ-YYYY-NNNN_MaterialName.pdf`. | Medium |

## 4.3 Vendor Management

| ID | Requirement | Priority |
|---|---|---|
| FR-VND-001 | Admin users shall be able to create, read, update, and deactivate vendor records. | High |
| FR-VND-002 | Vendor record shall include: Code, Name, Country, Contact Person, Email, Phone, and Active status. | High |
| FR-VND-003 | Inactive vendors shall not appear in any selection dropdowns across the system. | Medium |

## 4.4 Product and BOM Registry

| ID | Requirement | Priority |
|---|---|---|
| FR-BOM-001 | Admin users shall be able to create and manage Products and Raw Materials. | High |
| FR-BOM-002 | Admin users shall be able to define BOM entries linking raw materials to products with a standard quantity. | High |
| FR-BOM-003 | When a Product is selected in the PR form, the Raw Material dropdown shall filter to display only materials defined in that product's BOM. | Critical |

# 5. Non-Functional Requirements

## 5.1 Performance

| ID | Requirement |
|---|---|
| NFR-P-001 | Any application page shall load within 3 seconds under normal network conditions. |

| ID | Requirement |
|---|---|
| NFR-P-002 | PR form submission shall complete within 2 seconds. |
| NFR-P-003 | RFQ PDF generation and download initiation shall complete within 5 seconds. |
| NFR-P-004 | The system shall support up to 20 concurrent users without performance degradation. |

## 5.2 Security

| ID | Requirement |
|---|---|
| NFR-S-001 | All users shall authenticate with username and password before accessing any resource. |
| NFR-S-002 | Passwords shall be stored using bcrypt or Argon2 hashing. |
| NFR-S-003 | Role-based access control shall restrict operations based on user role. |
| NFR-S-004 | All user inputs shall be sanitized to prevent SQL injection and XSS attacks. |
| NFR-S-005 | User sessions shall expire after 60 minutes of inactivity. |
| NFR-S-006 | HTTPS shall be enforced in the production environment. |

## 5.3 Usability

| ID | Requirement |
|---|---|
| NFR-U-001 | A user with basic computer literacy shall be able to create a PR within 5 minutes without prior training. |
| NFR-U-002 | Required fields shall be visually distinguished from optional fields on all forms. |

| ID | Requirement |
|---|---|
| NFR-U-003 | Confirmation messages shall be displayed after successful PR creation and RFQ generation. |

## 5.4 Reliability and Maintainability

| ID | Requirement |
|---|---|
| NFR-R-001 | The system shall target 99% uptime during factory working hours (07:00–19:00). |
| NFR-R-002 | All database writes shall be transactional; no partial saves shall occur. |
| NFR-R-003 | Environment configurations shall be managed via environment variables and never hardcoded. |
| NFR-R-004 | The PDF template shall be a separate editable file to allow layout and branding changes without modifying application code. |

# 6. Database Design

## 6.1 Core Entities

| Entity | Description |
|---|---|
| `products` | Finished products manufactured at the factory |
| `raw_materials` | Raw materials used in production |
| `bom_lines` | Junction table linking products to their required raw materials |
| `users` | System users with assigned roles |
| `purchase_requests` | Digital Purchase Request records |
| `rfqs` | RFQ records linked to approved PRs |
| `vendors` | Vendor registry with contact information |

## 6.2 Key Relationships

| Relationship | Type | Description |
|---|---|---|
| Product → BOM Lines | 1 : M | One product has many BOM entries |
| Raw Material → BOM Lines | 1 : M | One material can appear in multiple products' BOMs |
| Product → Purchase Requests | 1 : M | One product can have many PRs |
| Raw Material → Purchase Requests | 1 : M | One material can be requested multiple times |
| Purchase Request → RFQ | 1 : 1 | One approved PR generates one RFQ |
| User → Purchase Requests | 1 : M | One user can create many PRs |

## 6.3 Entity Field Definitions

**`products`**

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| code | VARCHAR(20) | Unique, NOT NULL |
| name | VARCHAR(100) | NOT NULL |
| description | TEXT | Nullable |
| is_active | BOOLEAN | Default: true |
| created_at | TIMESTAMP | Default: NOW() |

**`raw_materials`**

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| code | VARCHAR(20) | Unique, NOT NULL |
| name | VARCHAR(100) | NOT NULL |

| Column | Type | Constraints |
|---|---|---|
| specification | TEXT | Nullable |
| default_uom | VARCHAR(20) | NOT NULL |
| is_active | BOOLEAN | Default: true |
| created_at | TIMESTAMP | Default: NOW() |

### `bom_lines`

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| product_id | INTEGER | FK → products, NOT NULL |
| raw_material_id | INTEGER | FK → raw_materials, NOT NULL |
| standard_qty | DECIMAL | NOT NULL |
| uom | VARCHAR | NOT NULL |

> Constraint: `UNIQUE (product_id, raw_material_id)`

### `users`

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| username | VARCHAR(50) | Unique, NOT NULL |
| full_name | VARCHAR(100) | NOT NULL |
| email | VARCHAR(150) | Unique, NOT NULL |
| password_hash | VARCHAR(255) | NOT NULL |
| role | ENUM | `planner` , `purchasing` , `manager` , `admin` |
| is_active | BOOLEAN | Default: true |
| created_at | TIMESTAMP | Default: NOW() |

### `purchase_requests`

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| pr_number | VARCHAR(20) | Unique, NOT NULL — format: `PR-YYYY-NNNN` |
| product_id | INTEGER | FK → products, NOT NULL |
| material_id | INTEGER | FK → raw_materials, NOT NULL |
| quantity | DECIMAL | NOT NULL, CHECK > 0 |
| uom | VARCHAR(20) | NOT NULL |
| required_date | DATE | NOT NULL |
| requester_id | INTEGER | FK → users, NOT NULL |
| department | VARCHAR(100) | Nullable |
| notes | TEXT | Nullable |
| status | ENUM | `draft` , `pending` , `approved` , `cancelled` |
| approved_by | INTEGER | FK → users, Nullable |
| approved_at | TIMESTAMP | Nullable |
| created_at | TIMESTAMP | Default: NOW() |
| updated_at | TIMESTAMP | Default: NOW() |

### `rfqs`

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| rfq_number | VARCHAR(20) | Unique, NOT NULL — format: `RFQ-YYYY-NNNN` |
| pr_id | INTEGER | FK → purchase_requests, Unique |
| issue_date | DATE | NOT NULL |
| due_date | DATE | NOT NULL |

| Column | Type | Constraints |
|---|---|---|
| pdf_file_path | VARCHAR(255) | Nullable |
| version | INTEGER | Default: 1 |
| generated_by | INTEGER | FK → users, NOT NULL |
| created_at | TIMESTAMP | Default: NOW() |

**`vendors`**

| Column | Type | Constraints |
|---|---|---|
| id | INTEGER | PK, Auto-increment |
| code | VARCHAR(20) | Unique, NOT NULL |
| name | VARCHAR(150) | NOT NULL |
| country | VARCHAR(80) | NOT NULL |
| contact_person | VARCHAR(100) | Nullable |
| email | VARCHAR(150) | Nullable |
| phone | VARCHAR(30) | Nullable |
| is_active | BOOLEAN | Default: true |
| created_at | TIMESTAMP | Default: NOW() |

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> **Prepared by:** Ghaith Hasan | SRS-PROC-MVP-001 v1.0 | 2026-03-16